

Admin Dashboard MVP - Wireframes & Specifications

Overview

The Admin Dashboard provides essential collection intelligence for Luciteria staff. This MVP focuses on three simple views: Overview Dashboard, User Collections List, and Demand Insights. No complex automation, triggers, or sync panels—just actionable data.

A. Admin Overview Dashboard

Route: `/app/admin`

Purpose

Single-page snapshot of collection activity across all users. Quick insights into what collectors own, want, and recent activity.

Wireframe

Luciteria Admin

[Overview]

[User Collections]

[Demand Insights]

[Back to Cabinet]

Collection Intelligence Hub

Last updated: May 27, 2026, 3:42 PM

[Refresh Data]

Total Users

247 collectors

Total Collections

18,943 items tracked

Avg Elements/User

76.7 elements

Active Users

89 (last 7d)

🏆 Top 10 Most Collected Elements

Rank	Element	Collections	% of Users	Trend (30d)
1	Carbon (C)	198	80.2%	↑ +12
2	Copper (Cu)	187	75.7%	↑ +8
3	Silicon (Si)	176	71.3%	↑ +15
4	Iron (Fe)	169	68.4%	↑ +7
5	Aluminum (Al)	162	65.6%	↑ +11
6	Sulfur (S)	154	62.3%	↑ +9
7	Zinc (Zn)	148	59.9%	→ +2
8	Nickel (Ni)	142	57.5%	↑ +6
9	Titanium (Ti)	139	56.3%	↑ +10
10	Magnesium (Mg)	135	54.7%	↑ +5

[Export to CSV]

⭐ Top 10 Most Wanted Elements

Rank	Element	Wishlist Cnt	Out of Stock?	Demand Score
1	Gallium (Ga)	89	⚠️ YES	🔥 High (89)
2	Bismuth (Bi)	76	✅ In Stock	🔥 High (76)
3	Selenium (Se)	68	⚠️ YES	🔥 High (68)
4	Antimony (Sb)	63	✅ In Stock	🔥 Medium (63)
5	Indium (In)	58	⚠️ YES	🔥 Medium (58)
6	Tellurium (Te)	52	✅ In Stock	🔥 Medium (52)
7	Cesium (Cs)	49	✅ In Stock	🔥 Medium (49)
8	Rhodium (Rh)	47	⚠️ YES	🔥 Medium (47)
9	Ruthenium (Ru)	44	✅ In Stock	🔥 Medium (44)
10	Hafnium (Hf)	41	⚠️ YES	🔥 Medium (41)

[Export to CSV]

📝 Recent Activity Feed

[View All]

Sarah K. added Iron (Fe) 50mm cube to collection

2 hours ago

Michael R. added Bismuth (Bi) ampoule to collection

3 hours ago

Jennifer L. added Gallium (Ga) 25.4mm to wishlist

4 hours ago

David P. completed "Transition Metals" milestone

5 hours ago

Emma W. added Selenium (Se) 10mm sample to collection

6 hours ago

- James T. added Copper (Cu) Lucite cube to collection 8 hours ago
- Olivia M. reached 50% collection completion 9 hours ago
- Noah H. added Titanium (Ti) 50mm cube to collection 11 hours ago
- Sophia G. added Carbon (C) ampoule to wishlist 13 hours ago
- Liam B. added Zinc (Zn) 25.4mm to collection 14 hours ago

Component Specifications

1. Key Stats Cards

Layout: 4 cards in a row (responsive: 2x2 on tablet, 1x4 on mobile)

Metric	Calculation	Update Frequency
Total Users	<code>COUNT(DISTINCT userId)</code> from <code>CollectionItem</code>	Real-time
Total Collections	<code>COUNT(*)</code> from <code>Collection-Item</code> where <code>state = 'owned'</code>	Real-time
Avg Elements/User	<code>Total Collections / Total Users</code>	Real-time
Active Users (7d)	<code>COUNT(DISTINCT userId)</code> from <code>CollectionItem</code> where <code>updatedAt > now() - 7 days</code>	Real-time

Visual Design:

- Large number (48px, bold)
- Descriptive label below (14px, gray)
- Subtle border, white background
- Hover effect: slight shadow

2. Top 10 Most Collected Elements

Data Source: `CollectionItem` table, `state = 'owned'`, grouped by `elementSymbol`

Columns:

- Rank (auto-numbered)
- Element (symbol + name, e.g., "Carbon (C)")
- Collections (count of owned items)
- % of Users ($\text{collections} / \text{total users} \times 100$)
- Trend (30d) - change in collections over last 30 days with arrow indicator

Sorting: Default by Collections descending

Export: CSV with columns: Rank, Element, Symbol, Atomic Number, Collections, Percentage, Trend 30d

3. Top 10 Most Wanted Elements

Data Source: CollectionItem table, state = 'wishlist', grouped by elementSymbol

Columns:

- Rank (auto-numbered)
- Element (symbol + name)
- Wishlist Count (count of wishlisted items)
- Out of Stock? - checks if any SKU for this element has `inventory = 0` in Product table
- Demand Score - visual indicator (High/Medium/Low) based on wishlist count

Demand Score Thresholds:

- 🔥 High: 50+ wishlists
- 🟡 Medium: 20-49 wishlists
- 🟢 Low: <20 wishlists

Export: CSV with columns: Rank, Element, Symbol, Atomic_Number, Wishlist_Count, Out_Of_Stock, Demand_Score

4. Recent Activity Feed

Data Source: `CollectionItem` table, joined with `User` and `Element` tables, ordered by `createdAt` DESC , limit 10

Entry Types:

- New collection addition: “{User} added {Element} {format} to collection”
- Wishlist addition: “{User} added {Element} to wishlist”
- Milestone: “{User} completed ‘{Milestone Name}’ milestone”
- Completion: “{User} reached {percentage}% collection completion”

Timestamp: Relative time (e.g., “2 hours ago”)

Interactions:

- Click user name → navigate to `/app/admin/users?userId={id}`
- Click element → show element details modal
- “View All” button → paginated full activity log

B. User Collections List

Route: `/app/admin/users`

Purpose

Browse all collectors, search by name/email, view individual collections, and export user data for analysis.

Wireframe

Luciteria Admin

[Overview]

[User Collections]

[Demand Insights]

[Back to Cabinet]

User Collections

Search by name or email...

[Search]

[Export All CSV]

Showing 247 collectors

User Name	Email	Elements	Complete	Last Active
Sarah Johnson	sarah@email.com	94	79.7%	2 hours ago
Michael Rodriguez	michael.r@email.com	88	74.6%	3 hours ago
Jennifer Lee	jlee@email.com	82	69.5%	1 day ago
David Park	dpark@email.com	76	64.4%	5 hours ago
Emma Wilson	emma.w@email.com	71	60.2%	6 hours ago
James Taylor	jtaylor@email.com	68	57.6%	2 days ago
Olivia Martinez	olivia.m@email.com	65	55.1%	9 hours ago
Noah Harris	noah.h@email.com	62	52.5%	11 hours ago
Sophia Garcia	sophia.g@email.com	58	49.2%	1 day ago
Liam Brown	lbrown@email.com	54	45.8%	14 hours ago
Ava Davis	ava.d@email.com	51	43.2%	3 days ago
Ethan Miller	ethan.m@email.com	48	40.7%	1 week ago
Isabella Anderson	isabella.a@email.com	45	38.1%	2 days ago
Mason Thomas	mason.t@email.com	42	35.6%	4 days ago
Mia Jackson	mia.j@email.com	39	33.1%	1 week ago

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User Detail Modal (Click any row)

Sarah Johnson's Collection[x Close]

Email: sarah@email.com

Shopify Customer ID: 5847392019

Collection Format: 25.4mm

Member Since: January 15, 2024

Elements Owned94

Completion79.7%

Wishlist Size12

Recent Activity:

Added Iron (Fe) 50mm cube - 2 hours ago

Added Bismuth (Bi) ampoule - 1 week ago

Completed "Transition Metals" milestone - 2 weeks ago

Top Wishlist Items:

1. Gallium (Ga) - Added 3 weeks ago

2. Selenium (Se) - Added 1 month ago

3. Antimony (Sb) - Added 2 months ago

[View Full Cabinet]

[Export User Data]

Component Specifications

1. Search Bar

Functionality:

- Real-time search as user types (debounced 300ms)
- Searches across: firstName, lastName, email
- Case-insensitive
- Clears with x button

No Results State:

No collectors found matching "xyz"
Try adjusting your search terms.

2. User Table

Data Source: User table joined with aggregated CollectionItem data

Columns:

Column	Source	Calculation
User Name	User.firstName + lastName	Direct
Email	User.email	Direct
Elements	Count of owned CollectionItems	COUNT(*) WHERE state = 'owned'
Complete	Percentage completion	(Elements / 118) × 100

| Last Active | Most recent CollectionItem update | `MAX(CollectionItem.updatedAt)` formatted as relative time |

Sorting:

- Default: Last Active descending (most recent first)
- Clickable column headers to sort by any column
- Visual indicator (↑ ↓) on sorted column

Pagination:

- 15 rows per page
- Standard pagination controls
- Shows total count: "Showing 247 collectors"

Row Interaction:

- Click anywhere on row → open User Detail Modal
- Hover effect: light background color change

3. Export All CSV

Button: Top-right of page

Exported Data:

```
User_ID,Name,Email,Shopify_Customer_ID,Elements_Owned,Completion_Percent,Wishlist_Size,Last_Active,Me
12345,Sarah Johnson,sarah@email.com,5847392019,94,79.7,12,2026-05-27T13:42:00Z,2024-01
-15
...
```

Filename: `luciteria_collectors_export_{timestamp}.csv`

4. User Detail Modal

Trigger: Click table row

Data Displayed:

- User profile info (email, Shopify ID, format, join date)
- Collection stats (owned count, completion %, wishlist size)
- Recent activity (last 5 actions)
- Top wishlist items (top 5 by priority)

Actions:

- **View Full Cabinet:** Opens user's cabinet view (read-only for admin)
- **Export User Data:** Downloads CSV with user's complete collection
- **Close (×):** Dismisses modal

Read-Only Cabinet View:

- Navigates to `/app/admin/users/{userId}/cabinet`
 - Shows user's periodic table with their owned/wishlist states
 - Banner at top: "Viewing {User Name}'s Cabinet (Read-Only)"
 - No edit capabilities for admin
-

C. Demand Insights Panel

Route: `/app/admin/demand`

Purpose

Identify restocking opportunities and high-demand items to inform inventory purchasing decisions.

Wireframe

Luciteria Admin

[Overview]

[User Collections]

[Demand Insights]

[Back to Cabinet]

Demand Insights

Inventory intelligence **for** restocking decisions

Out-of-Stock Watchlist

Items currently unavailable with active collector demand

SKU	Element	Format	Waitlist Cnt	Days Out Stock
GA-25-PU	Gallium (Ga)	25.4mm	89	47 days
SE-10-AM	Selenium(Se)	10mm	68	32 days
IN-50-CU	Indium (In)	50mm cube	58	18 days
RH-AM-PU	Rhodium (Rh)	Ampoule	47	61 days
HF-25-PU	Hafnium (Hf)	25.4mm	41	22 days
OS-AM-PU	Osmium (Os)	Ampoule	38	89 days
IR-10-PU	Iridium (Ir)	10mm	35	76 days
TA-25-CU	Tantalum(Ta)	25.4mm	31	15 days
RE-AM-PU	Rhenium (Re)	Ampoule	28	54 days
TL-10-AM	Thallium(Tl)	10mm ampoule	24	19 days

[Export **for** Restocking]

[View All Out-of-Stock]

High-Demand Items

Most wanted by collectors (**in** stock + **out** of stock)

SKU	Element	Wishlist Cnt	Missing From	Stock Status
GA-25-PU	Gallium (Ga)	89	158 users	Out of Stock
BI-50-CU	Bismuth (Bi)	76	171 users	In Stock (45)
SE-10-AM	Selenium(Se)	68	179 users	Out of Stock
SB-25-PU	Antimony(Sb)	63	184 users	In Stock (28)
IN-50-CU	Indium (In)	58	189 users	Out of Stock
TE-AM-PU	Tellurium	52	195 users	In Stock (19)
CS-AM-GL	Cesium (Cs)	49	198 users	In Stock (12)
RH-AM-PU	Rhodium (Rh)	47	200 users	Out of Stock
RU-10-PU	Ruthenium	44	203 users	In Stock (34)
HF-25-PU	Hafnium (Hf)	41	206 users	Out of Stock

[Export High-Demand List]

[Sort by: Wishlist Count]

Component Specifications

1. Out-of-Stock Watchlist

Purpose: Prioritize restocking based on active collector demand

Data Source:

- Product table where `inventory = 0` OR `availableForSale = false`
- Joined with `CollectionItem` where `state = 'wishlist'`
- Calculate days out of stock: `NOW() - lastInStockDate`

Columns:

Column	Source	Notes
SKU	Product.sku	Direct
Element	Product.title parsed	Extract element name + symbol
Format	Product.variant or tags	10mm, 25.4mm, 50mm, Lucite Cube, Ampoule
Waitlist Cnt	COUNT(CollectionItem) WHERE state = 'wishlist'	Number of collectors wanting this SKU
Days Out Stock	NOW() - Product.lastInStockDate	Visual indicator based on severity

Days Out of Stock Color Coding:

- ● Red: 30+ days (critical)
- ● Yellow: 15-29 days (moderate)
- ● Green: <15 days (recent)

Sorting: Default by Waitlist Count descending

Export Button: "Export for Restocking"

- Generates CSV: `restocking_priority_{timestamp}.csv`
- Columns: SKU, Element, Atomic_Number, Format, Waitlist_Count, Days_Out_Of_Stock, Supplier_Info (if available)
- Sorted by demand priority

2. High-Demand Items

Purpose: Identify popular elements regardless of stock status (helps with inventory planning)

Data Source:

- All SKUs with active wishlists (regardless of stock status)
- Aggregated wishlist counts
- "Missing From" = collectors who don't own this element yet

Columns:

Column	Source	Calculation
SKU	Product.sku	Direct
Element	Product.title	Parse element name
Wishlist Cnt	COUNT(CollectionItem) WHERE state = 'wishlist'	Total wishlists
Missing From	COUNT(Users) WHERE element NOT in owned collection	Potential market size
Stock Status	Product.inventory	"In Stock (qty)" or "Out of Stock"

Stock Status Visual:

- ✔ Green: In Stock (shows quantity)
- ⚠ Red: Out of Stock

Sorting Options (dropdown):

- Wishlist Count (default)
- Missing From (potential market)
- Stock Status (show out-of-stock first)
- Element (alphabetical)

Export Button: “Export High-Demand List”

- Generates CSV: `high_demand_elements_{timestamp}.csv`
- Includes all columns plus: `Current_Inventory`, `Price`, `Last_Restock_Date`

3. Visual Indicators**Demand Severity Badges:**

-  Critical (80+ wishlists)
-  High (50-79 wishlists)
-  Medium (20-49 wishlists)
- Low (<20 wishlists)

Stock Status Icons:

-  In Stock (quantity shown)
-  Out of Stock
-  Restocking Soon (**if** flag set in admin)

Technical Requirements

Data Refresh Strategy

- **Real-time:** User-facing metrics update on page load
- **Cached:** Aggregate stats cached for 5 minutes (reduce DB load)
- **Manual Refresh:** “Refresh Data” button forces cache invalidation

Performance Targets

- Dashboard load: <2 seconds
- User list search: <500ms
- Table sorting: <200ms (client-side after initial load)
- Export generation: <5 seconds for full dataset

Access Control

- All admin routes require `User.isStaff = true`
- If non-staff user attempts access: redirect to `/app/cabinet` with toast error
- Audit log: Track all admin actions (exports, user views)

Mobile Responsiveness

- **Desktop** (>1200px): Full layout as shown
- **Tablet** (768-1199px): Stack tables vertically, reduce columns
- **Mobile** (<768px): Card-based layout instead of tables

Browser Support

- Chrome/Edge 90+
 - Firefox 88+
 - Safari 14+
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Key Design Decisions

1. MVP Scope

What's Included:

- Essential collection intelligence (who owns what, what's wanted)
- Restocking decision support
- Basic user management (view/search)

What's Deferred (Post-MVP):

- Automated notification triggers UI
- Bulk operations (mass email, bulk updates)
- Advanced analytics (trend charts, forecasting)
- Subscription management integration
- Real-time activity stream

2. Data Export Over Automation

- MVP focuses on exporting data for manual decisions
- No automated restocking triggers, notification campaigns, or email blasts
- Staff downloads CSVs and uses existing tools (Shopify, email client, Excel)
- Reduces complexity, faster to ship

3. Read-Only User Cabinet View

- Admins can view any user's collection but NOT edit
- Prevents accidental data corruption
- If editing needed, staff uses Shopify admin tools

4. Simple Tables, No Charts

- MVP uses sortable tables, not graphs/charts
 - Reduces development time (no charting library)
 - Tables are more actionable (can scan and copy SKUs)
 - Phase 3 can add visualizations if needed
-

Success Metrics

For Staff

- **Time saved:** Restocking decisions made in <10 minutes (vs. manual Shopify queries)
- **User support:** Find user collections in <30 seconds
- **Data export:** All key reports exportable in 1 click

For Business

- **Demand visibility:** Out-of-stock items with waitlists identified immediately
 - **Inventory optimization:** Restock based on actual collector demand, not guesses
 - **User engagement insights:** Track collection growth and completion rates
-

Summary

Total Views: 3 (Overview, User Collections, Demand Insights)

Total Components: 9 (4 cards, 4 tables, 1 activity feed)

Export Options: 4 (Overview elements, user list, restocking priority, high-demand)

Philosophy: Simple, actionable intelligence. Staff can see what collectors own/want, identify restocking priorities, and export data for decisions—all without complexity or automation overhead.